COMPLETE PRACTICAL CAD HANDBOOK • REVISION 2021

Engineering Graphics Using CAD Software

A full laboratory handbook for learning CAD tools, precision drafting, six foundational figures, four isometric drawings, two complete 2D engineering drawings, plotting, lab examination workflow and open-ended project reporting.

Course Code
2038

Programme
Electrical & Electronics / Renewable Energy / Electrical & EV Technology

Semester
2

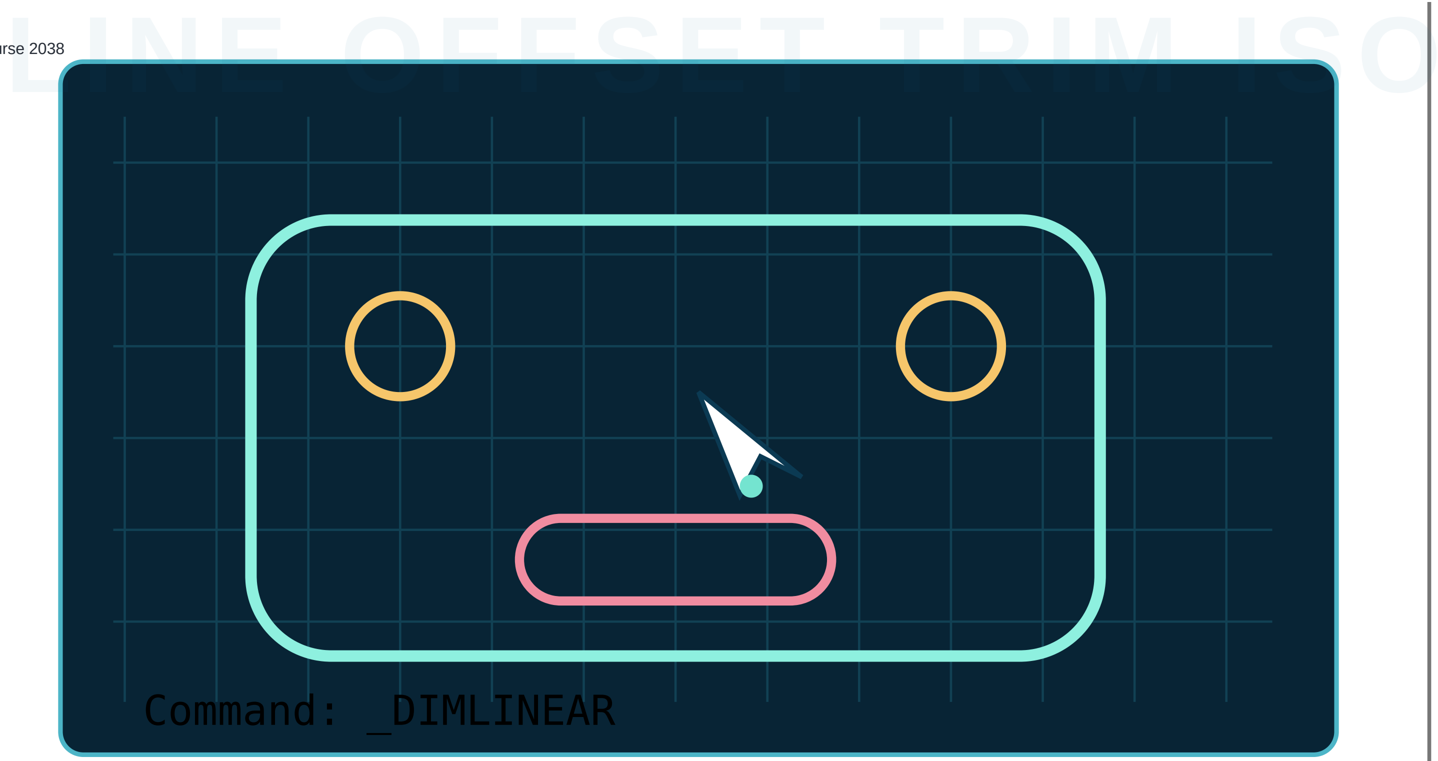
Credits
No Credit

Category
Engineering Science

Periods
3 practical/week • 45/semester

How this handbook matches the official syllabus

The official syllabus specifies basic CAD tools, at least six CAD figures, at least four isometric figures, at least two complete two-dimensional figures, two three-hour lab examinations and a 15-mark open-ended group activity. Every one of those requirements is built into this handbook. Verified again against the uploaded three-page official syllabus: programme, no-credit status, 45 practical periods, CO-PO mapping, lab-exam split, references, online resources and the open-ended experiment requirement are reflected here.



Official practical path

Identify tools and apply them to sketches.

Develop at least six CAD figures.

Develop at least four isometric figures.

Construct at least two complete 2D figures.

WHY THIS COURSE MATTERS

CAD converts engineering ideas into precise, editable drawings

Manual drawing develops visualisation; CAD adds speed, accuracy, revision control, layers, reusable geometry and professional plotting. Electrical and renewable-energy students use CAD for layouts, mounting plates, panel drawings, cable routes, component details and equipment arrangement.

Accuracy

Coordinates, object snaps and constraints reduce measurement and alignment errors.

Editability

MOVE, COPY, OFFSET, TRIM and grips modify geometry without redrawing the entire sheet.

Course 2038

Communication

Layers, dimensions, line types, text and plotting turn geometry into an engineering document.

How to use this handbook

1. Set up

Study units, workspace, coordinates, layers and object snaps before drawing.

2. Observe

Follow the command sequence and the geometric reason behind each step.

3. Draw

Complete every minimum syllabus figure and save each file with a clear name.

4. Verify

Check dimensions, symmetry, layer use, line type, title block and plot preview.

Lab exam habit

Read the complete question first. Plan layers and construction lines, save immediately, then draw. Use object snaps instead of visual guessing. Dimension only after geometry is complete and verified.

CLICKABLE CONTENTS

Open any handbook section

OFFICIAL SYLLABUS ANALYSIS

Course identity, objectives and prerequisites

Programme	Diploma in Electrical and Electronics Engineering / Renewable Energy / Electrical Engineering & Electric Vehicle Technology	**Semester**	2
Course Code	2038	**Credits**	No Credit
Official Course Title	Engineering Graphic using CAD software	**Course Category**	Engineering Science
Periods per week	3 (L:0, T:0, P:3)	**Periods per semester**	45

Course Objective 1

To use CAD software for drafting and modelling.

Student meaning: operate the interface, choose tools correctly, create precise geometry and edit drawings efficiently.

Course Objective 2

To learn the basics of 2D modelling of engineering components.

Student meaning: convert dimensions and geometry into a complete, readable engineering drawing.

Prerequisites

Topic	Course name	Semester/level
Basics of engineering graphics	Engineering Graphics	Semester 1
Geometry	Mathematics	Secondary School

COURSE OUTCOMES

What the student must demonstrate

CO	Official outcome	Hours	Cognitive level	Evidence in this handbook
CO1	Identify the tools in CAD software.	9	Applying	Interface, setup, coordinates, precision, draw/modify/annotation tools.
CO2	Develop and draw figures using CAD software.	12	Applying	Six dimensioned 2D construction exercises.
CO3	Sketch and practice isometric drawings using CAD software.	12	Applying	Four isometric exercises and isoplane workflow.
CO4	Construct two dimensional figures using CAD software.	6	Applying	Two complete engineering drawing sheets.
—	Lab Exam	6	—	Two three-hour model lab examinations.

CO-PO mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	Interpretation
CO1	—	—	3	—	—	—	—	Strongly mapped to PO3.
CO2	3	—	—	3	—	—	—	Strongly mapped to PO1 and PO4.
CO3	3	—	—	3	—	—	—	Strongly mapped to PO1 and PO4.
CO4	3	—	—	—	—	—	—	Strongly mapped to PO1.

Mapping scale

3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped.

MODULE-WISE ROADMAP

Official hours and minimum drawing count

Module outcome	Description	Hours	Level	Minimum evidence
M1.01	Study basic tools of CAD software and drawings.	3	Understanding	Interface and tool identification record.
M1.02	Apply software tools and draw the sketches.	6	Applying	Precision setup and guided sketches.
M2.01	Develop and draw figures using CAD.	12	Applying	At least 6 figures.
Lab Exam 1	Practical examination after Module 2.	3	—	One complete precision drawing task.
M3.01	Develop and draw isometric figures using CAD.	12	Applying	At least 4 figures.
M4.01	Develop and draw two dimensional figures using CAD.	6	Applying	At least 2 figures.
Lab Exam 2	Practical examination after Module 4.	3	—	Isometric/2D complete drawing task.

Hour check

9 + 12 + 3 + 12 + 6 + 3 = 45 periods, exactly matching the semester allocation.

MODULE 1 • CO1 • 9 HOURS

CAD Tools, Workspace and Precision Drafting

M1.01: Study basic tools and drawings. M1.02: Apply software tools and draw sketches.

1. CAD workspace

Drawing area

The model space where geometry is created at full scale.

Command line

Shows prompts, options and entered values. Read it continuously.

Ribbon/toolbars

Groups Draw, Modify, Annotation, Layers and Properties tools.

Status bar

Controls GRID, SNAP, ORTHO, POLAR, OSNAP, OTRACK and isometric drafting.

Model and layout

Model space holds full-size geometry; layout space prepares paper and plot scale.

Navigation

ZOOM and PAN change the view without changing object dimensions.

Malayalam Support: Model space-ൽ object actual size-ൽ draw ചെയ്യുക. Zoom/Pan view മാത്രം മാറ്റും; drawing size മാറ്റില്ല.

2. Start every drawing correctly

1. Create a new drawing using a suitable metric template.
2. Run `UNITS`; choose Decimal and millimetres where required.
3. Save immediately with a meaningful name such as `2038_M2_Figure01.dwg`.
4. Create layers for OBJECT, CENTRE, HIDDEN, DIMENSION, CONSTRUCTION and TEXT.
5. Set OSNAP modes: Endpoint, Midpoint, Centre, Intersection, Quadrant and Tangent as needed.
6. Use ZOOM Extents after major stages and save repeatedly.

3. Coordinate entry

Method	Syntax	Use	Example
Absolute Cartesian	x,y	Point measured from origin.	50,30
Relative Cartesian	@Δx,Δy	Point measured from previous point.	@80,0
Relative polar	@distance<angle	Length and direction from previous point.	@60<30
Direct distance	Point cursor + type length	Fast orthogonal/polar drawing.	Move right, type 100

Malayalam Support: @ ചേർത്ത coordinate previous point-നെ reference ആക്കും. @50,0 എന്നത് നിലവിലെ point-ൽ നിന്ന് 50 mm right എന്നർത്ഥം.

4. Precision controls

ORTHO

Restricts cursor movement to horizontal and vertical directions.

POLAR

Tracks selected angles such as 30°, 45° or 90°.

OSNAP

Locks the cursor to exact geometric points; never estimate visually.

Object tracking

Projects temporary alignment paths from snapped points.

Common mistake
Clicking near an endpoint is not the same as snapping to it. Confirm the OSNAP marker and tooltip before clicking.

5. Core draw and modify tools

Group	Commands	Purpose
Draw	LINE, PLINE, CIRCLE, ARC, RECTANGLE, POLYGON, ELLIPSE	Create primary geometry.
Modify	MOVE, COPY, ROTATE, MIRROR, OFFSET, TRIM, EXTEND, FILLET, CHAMFER, SCALE, STRETCH, ARRAY	Edit and repeat geometry.
Properties	LAYER, MATCHPROP, PROPERTIES	Control colour, line type, line weight and organisation.
Annotation	TEXT/MTEXT, DIM, MLEADER, HATCH	Add dimensions, notes, leaders and section patterns.
View/file	ZOOM, PAN, SAVE, SAVEAS, PLOT	Navigate, protect and output the work.

6. Drawing standards refresher

Layer	Suggested line type	Typical use
OBJECT	Continuous, thick	Visible outlines.
CENTRE	Centre line, thin	Axes and centres of circles.
HIDDEN	Dashed, thin	Invisible edges.
DIMENSION	Continuous, thin	Dimension and extension lines.
CONSTRUCTION	Continuous, very thin	Temporary layout geometry.
TEXT	Continuous	Notes and title block.

7. Guided starter sketch: centred rectangle with holes

Plate	Hole centres	Hole size	Tools
100 × 60	15 from adjacent edges	Ø10	RECTANGLE, OFFSET, CIRCLE, COPY

Suggested sequence: RECTANGLE → OFFSET 15 from four sides → CIRCLE R5 at first intersection → COPY to other intersections → erase/trim construction lines → DIM.

Module 1 practice questions

VERY SHORT

Differentiate model space and layout space.

COMMAND

Write the relative polar coordinate entry for a 75 mm line at 30°.

APPLICATION

List the OSNAP modes needed to draw a line tangent to two circles.

PROCEDURE

Write the correct startup procedure for a metric engineering drawing.

ERROR ANALYSIS

A drawing looks aligned but gaps appear when zoomed. Identify likely causes and corrections.

PRACTICAL

Create the 100 × 60 plate, apply dimensions and save it with the specified file name.

MODULE 2 • CO2 • 12 HOURS

Six Foundational CAD Figures

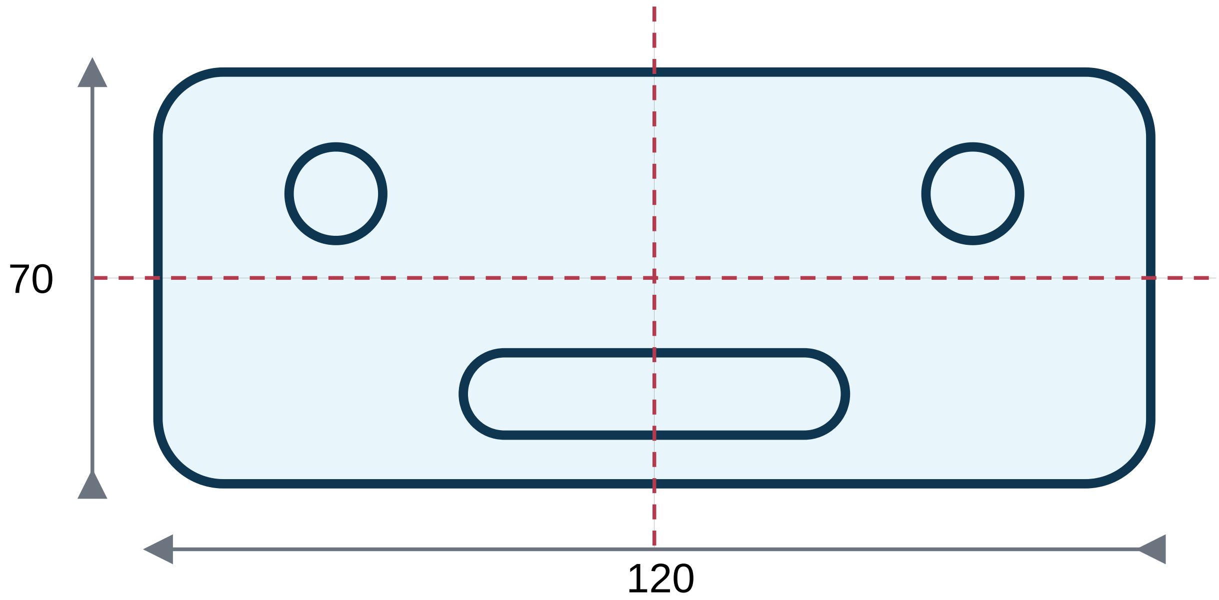
The official syllabus requires at least six figures. The following exercises progress from simple offsets to tangent geometry, arrays and complete dimensioning.

Overall
120 × 70

Corner radius
R10

Holes
2 × Ø12

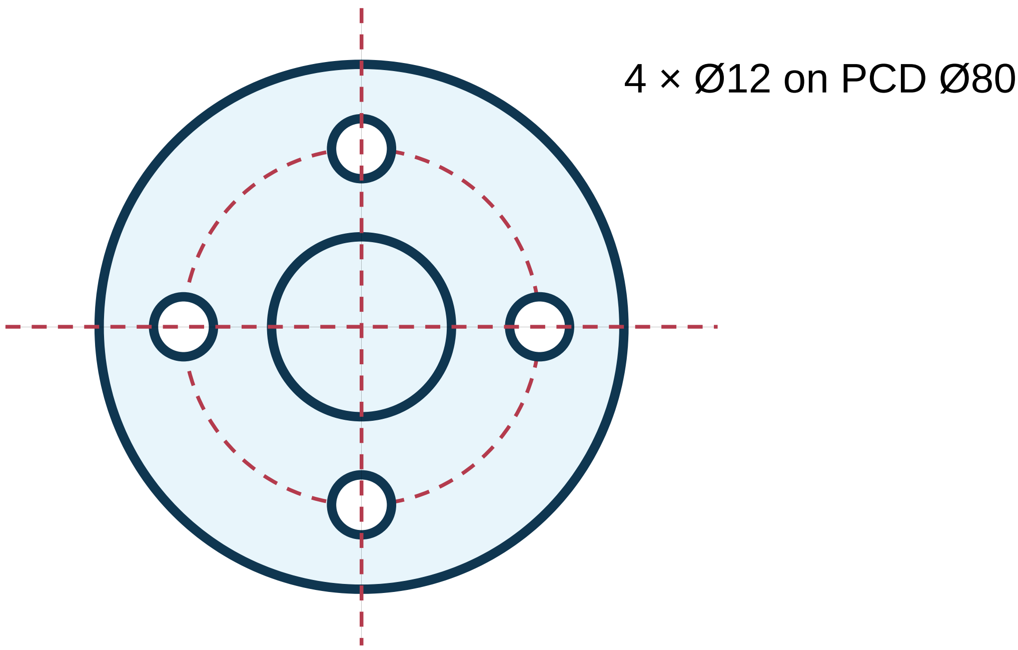
Slot
50 × 14



Suggested construction: outer rectangle → fillet corners → centre lines → two holes → slot from two circles and tangent lines → trim → dimension.

1. Draw a 120 × 70 rectangle.
2. FILLET all corners with radius 10.
3. Create horizontal and vertical centre lines.
4. Locate hole centres and draw Ø12 circles.
5. Create the slot using two R7 circles 50 mm centre-to-centre and tangent lines; TRIM inner arcs.
6. Apply dimensions and centre marks.

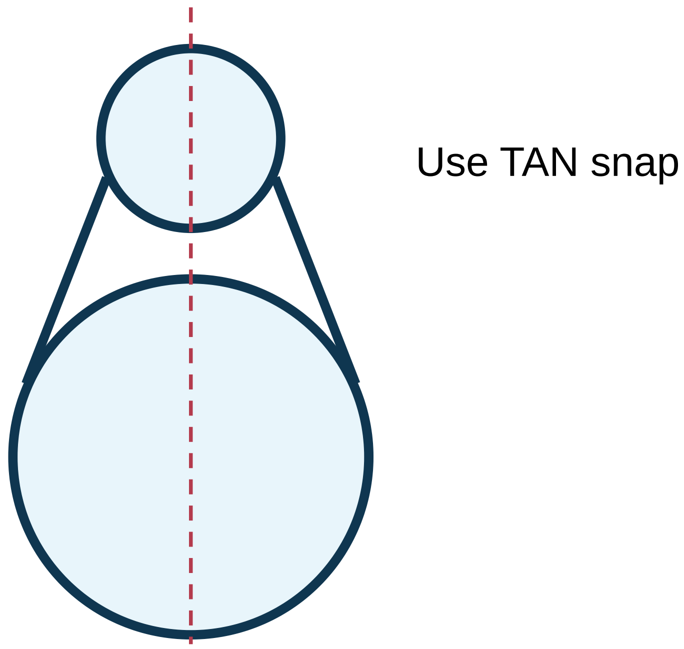
Outer Ø120	**Bore** Ø40	**Pitch circle** Ø80	**Bolt holes** 4 × Ø12



Draw one bolt hole on the pitch circle, then create a four-item polar array around the flange centre.

1. Draw concentric Ø120, Ø80 construction and Ø40 circles.
2. Place one Ø12 hole at the top quadrant of the pitch circle.
3. Use ARRAYPOLAR: centre = flange centre, items = 4, fill angle = 360°.
4. Place centre marks and diameter dimensions.

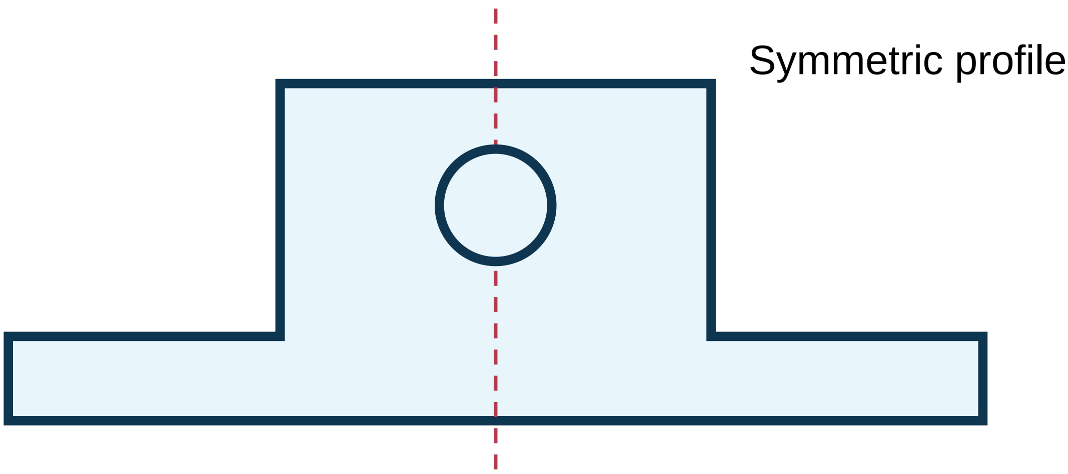
Base circle Ø60	**Top circle** Ø30	**Centre distance** 70 vertical	**Sides** Common tangents



The profile is built from two circles joined by common external tangents; unwanted arcs are trimmed.

1. Draw the two circles on a common vertical centre line.
2. Use LINE with Tangent OSNAP from the small circle to the large circle on each side.
3. TRIM internal circle portions and retain a closed cam outline.
4. Add centre marks and dimensions.

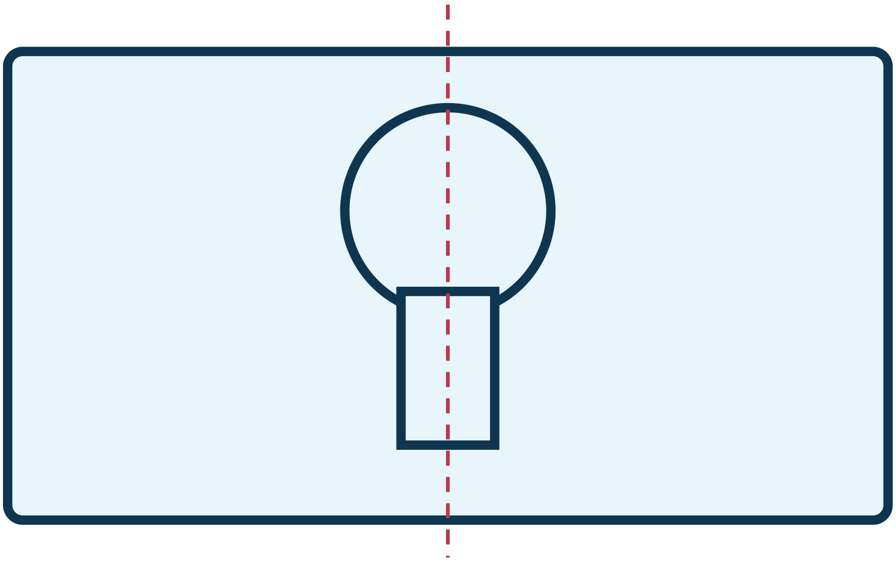
Base	**Web**	**Top**	**Corner**
140 × 20	60 high	80 × 20	R8 / 10×10 chamfer



Draw half of the symmetric outline accurately and MIRROR about the vertical centre line.

1. Draw the base and vertical centre line.
2. Use PLINE for one half of the stepped outline.
3. Apply required fillet/chamfer.
4. MIRROR the completed half, then draw the central hole.

Plate	**Upper circle**	**Stem**	**Position**
100 × 80	Ø30	14 wide × 30	Centred



Create the circular opening and rectangular stem, then TRIM their overlapping boundary to form one keyhole opening.

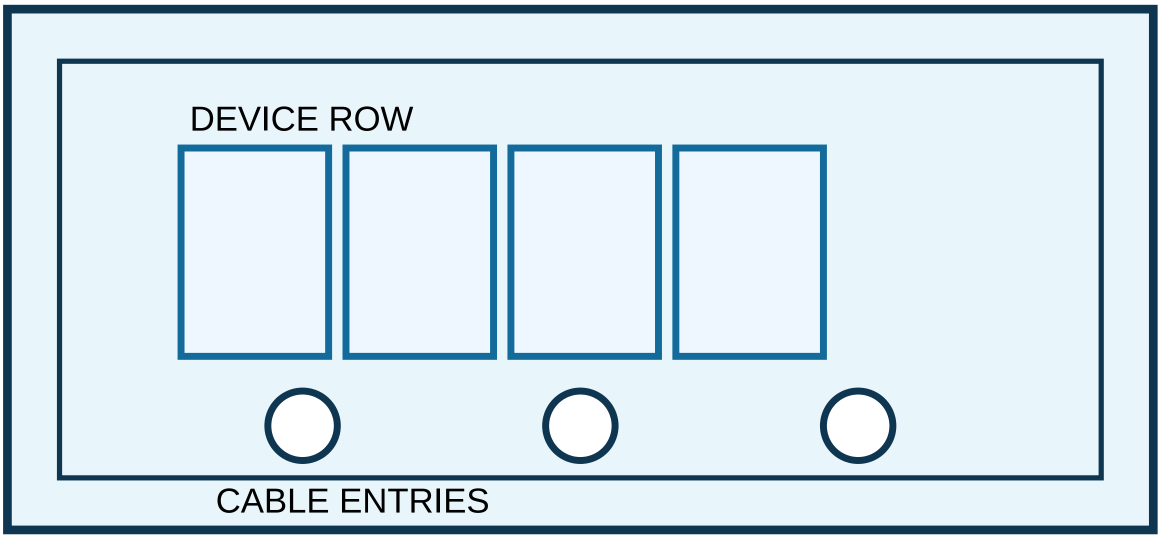
1. Draw the outer plate and centre lines.
2. Draw Ø30 at the specified centre.
3. Draw a 14 mm wide centred stem below the circle.
4. TRIM the common circle segment and internal top line of the stem.

Panel
180 × 120

Border
10 internal

DIN devices
4 modules 25 × 50

Cable entry
3 × Ø20



A practical layout exercise combining borders, repeated components, holes, labels and dimensions.

1. Draw panel outline and OFFSET 10 for the inner border.
2. Draw one device rectangle, then COPY or ARRAYRECT for four modules.
3. Draw one Ø20 cable entry and array/copy to three positions.
4. Add labels and overall/spacing dimensions on separate layers.

Module 2 completion evidence
Submit six separate DWG files or one organised workbook drawing containing all six figures, with correct layers, dimensions and file names.

Module 2 practice questions

PROCEDURE
Write a command sequence for the four-hole flange.

ERROR ANALYSIS
A polar array rotates the hole incorrectly. Explain which array settings must be checked.

GEOMETRY
Explain why Tangent OSNAP is essential for Figure 3.

PRACTICAL
Create any one of the six figures from dimensions without tracing the sample.

EDITING
Compare OFFSET, COPY and ARRAY for repeated geometry.

PLOT
Prepare Figure 6 for A4 plotting with readable dimensions.

MODULE 3 • CO3 • 12 HOURS

Isometric Drawing Using CAD

The official syllabus requires at least four isometric figures. These exercises build from a box to an engineering machine outline.

1. Isometric drafting essentials

Isometric axes

One vertical axis and two receding axes at 30° to the horizontal.

Isoplane

Left, Top and Right isoplanes control cursor orientation and isocircle plane.

Isocircle

Circles appear as ellipses; use ELLIPSE → Isocircle in isometric snap mode.

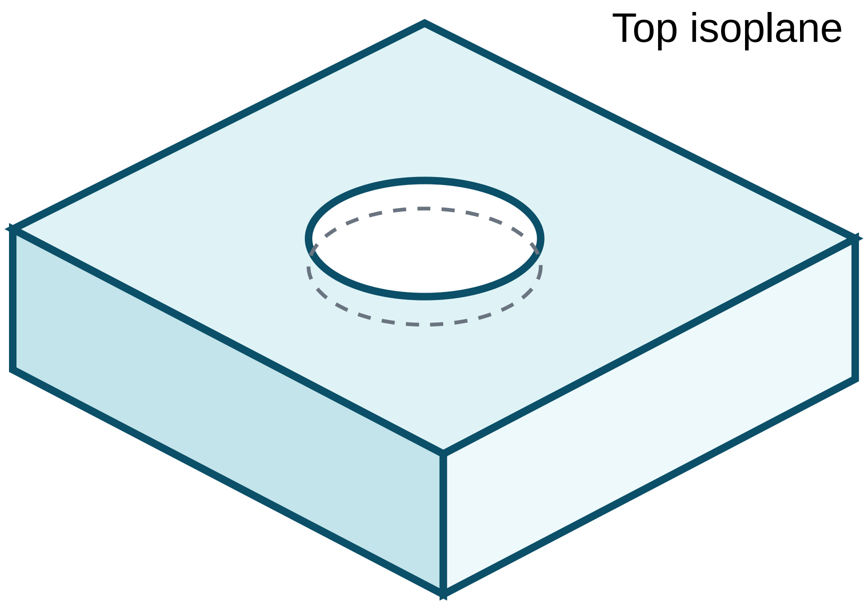
Typical setup: SNAP → Style → Isometric | toggle isoplane with **F5** or **Ctrl+E** | use ORTHO and POLAR.

Malayalam Support: Isometric mode-ൽ F5 ഉപയോഗിച്ച് Left/Top/Right isoplane മാറ്റാം. Circle draw ചെയ്യാൻ normal CIRCLE അല്ല, ELLIPSE → Isocircle ഉപയോഗിക്കുക.

Block
80 × 50 × 40

Hole
Ø20 through top

Tools
LINE, OFFSET, ELLIPSE Isocircle, TRIM



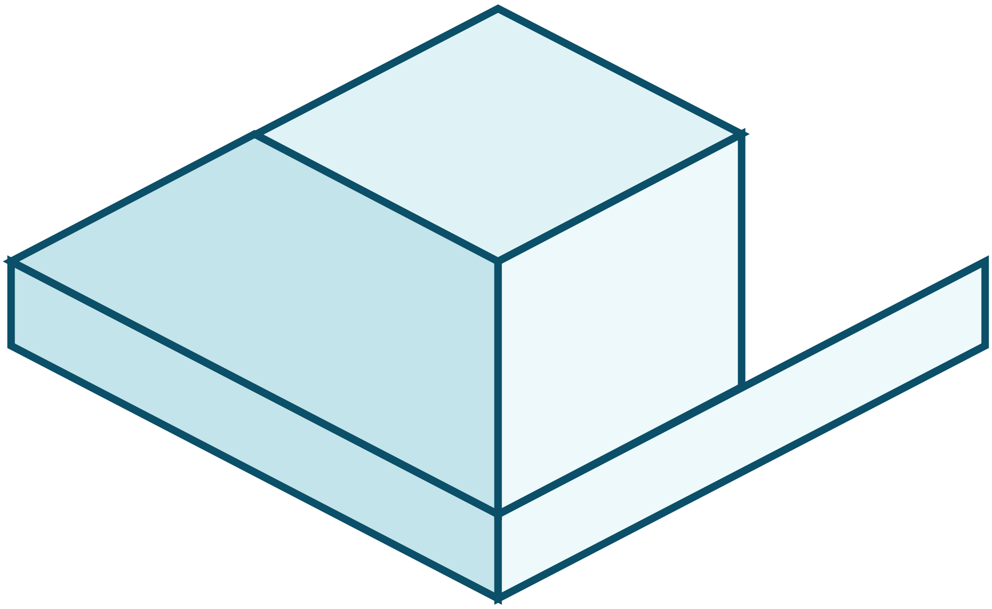
Construct the isometric box first. Locate the hole centre on the top face and create an isocircle.

1. Draw the three isometric axes from one corner.
2. Create the 80 × 50 × 40 bounding box.
3. Locate the top-face centre with diagonals/construction offsets.
4. Select Top isoplane and create Ø20 isocircle.
5. Add hidden lower ellipse only if the drawing convention/instructor requires it.

Overall
90 × 60 × 60

Step
45 length × 30 height

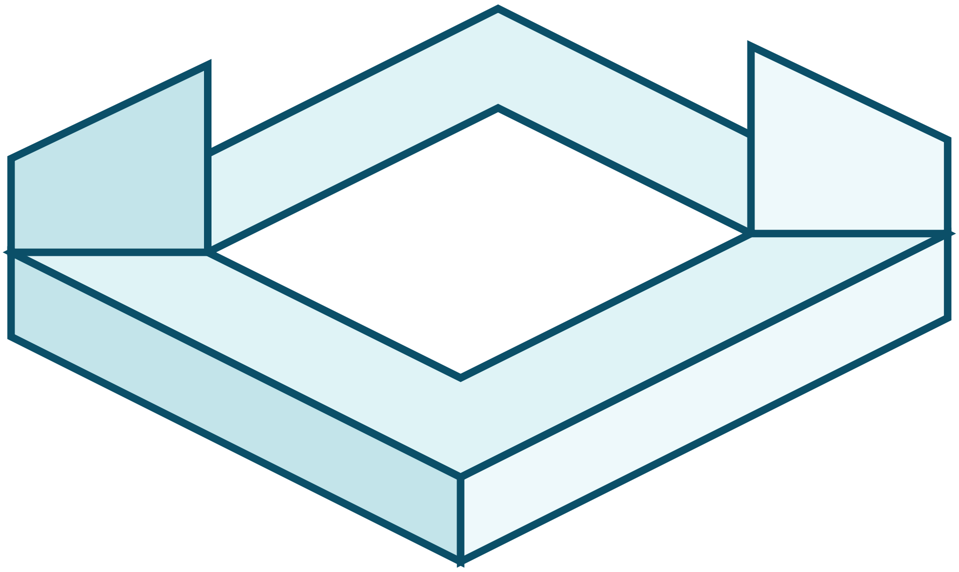
Tools
LINE, OFFSET, TRIM



Use an overall box, then remove the upper front portion conceptually. Construction-box thinking reduces errors.

1. Draw the complete 90 × 60 × 60 bounding box lightly.
2. Mark the step length and height on the correct axes.
3. Draw the new visible edges and TRIM construction edges.
4. Check that all receding lines remain parallel to the isometric axes.

Outer	Wall	Base	Tools
90 × 50 × 60	12	12	LINE, OFFSET, COPY, TRIM

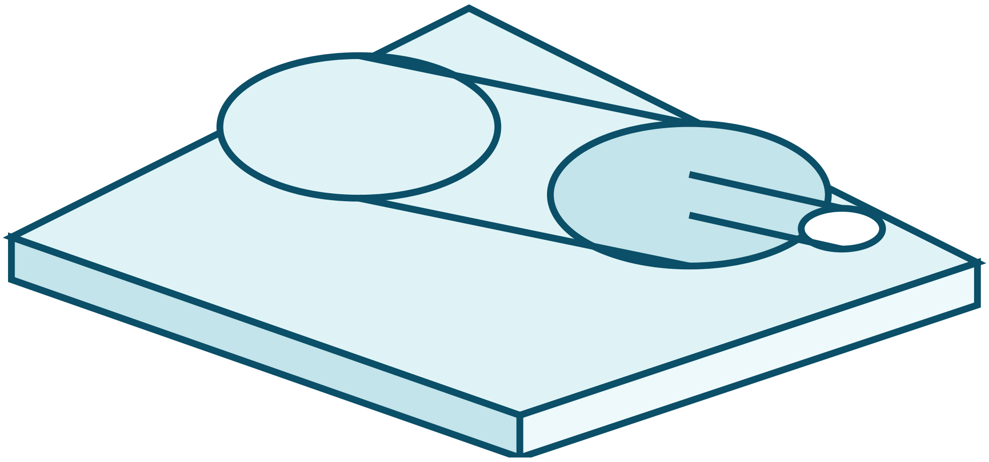


Build the base slab, then two equal side walls. Use offsets to maintain constant wall thickness.

1. Draw the base plate as an isometric slab.
2. OFFSET/construct 12 mm wall thickness on left and right.
3. Raise both walls to 60 mm height.
4. Remove hidden/construction geometry and verify symmetry.

Body	**Shaft**	**Base**	**Tools**
Ø60 × 100	Ø16 × 35	120 × 55 × 10	ELLIPSE Isocircle, LINE, OFFSET, TRIM

Simplified training form



A simplified training model, not a manufacturing drawing. It integrates isocircles, cylindrical form, shaft and base.

1. Draw the base plate.
2. Use the correct side isoplane to draw the motor-body end isocircles.
3. Join corresponding tangent points to form the cylinder.
4. Repeat for the shaft and add simple mounting feet.
5. Clean up hidden overlaps and add a descriptive title.

Isometric common mistakes

Using normal circles; changing to the wrong isoplane; measuring along the screen instead of isometric axes; allowing receding lines to drift away from 30°; trimming visible edges needed for the final form.

Module 3 practice questions

VERY SHORT

Name the three isoplanes.

COMMAND

Explain how to draw a circle on an isometric top face.

APPLICATION

Write a procedure for creating a cylindrical object in isometric CAD.

PRACTICAL

Draw any one of the four isometric exercises from dimensions.

ERROR ANALYSIS

An isocircle appears on the wrong plane. State the correction.

VISUALISATION

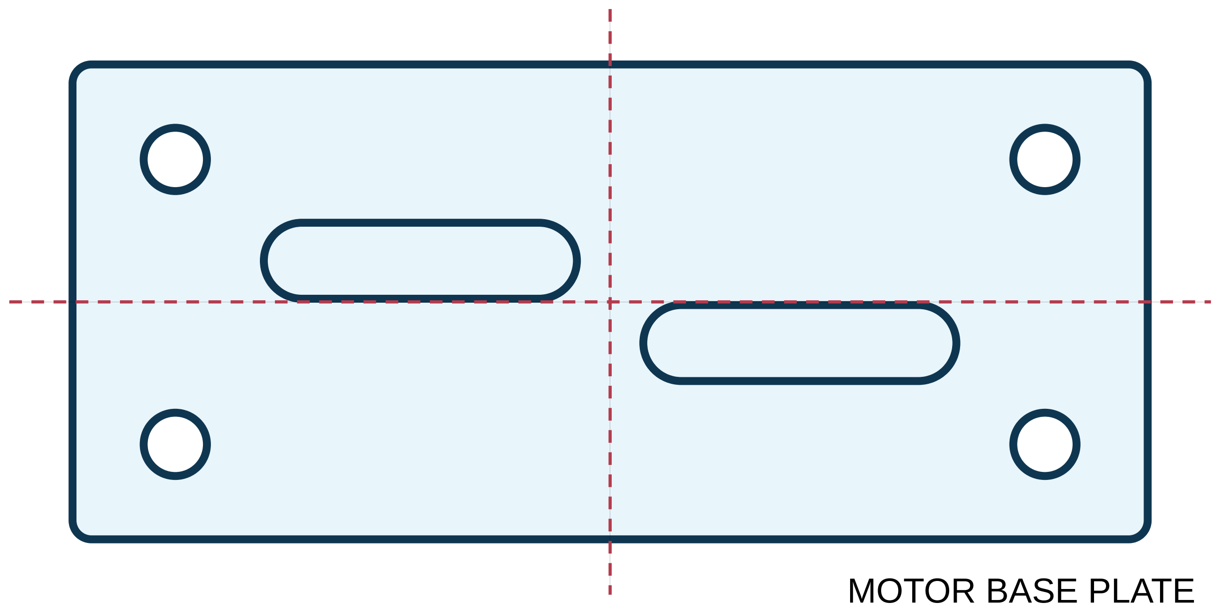
Describe how a bounding box helps create a stepped block.

MODULE 4 • CO4 • 6 HOURS

Two Complete Two-Dimensional Engineering Drawings

The official syllabus requires at least two figures. Unlike the shorter Module 2 exercises, these are complete sheets with layers, annotation, title, dimensions and plot preparation.

Overall	Corner holes	Motor slots	Centre distance
200 × 120	4 × Ø14, 20 from edges	2 × 70 × 16	80 between slots



A complete plate drawing with visible outlines, centre lines, holes, slots, dimensions, note and title.

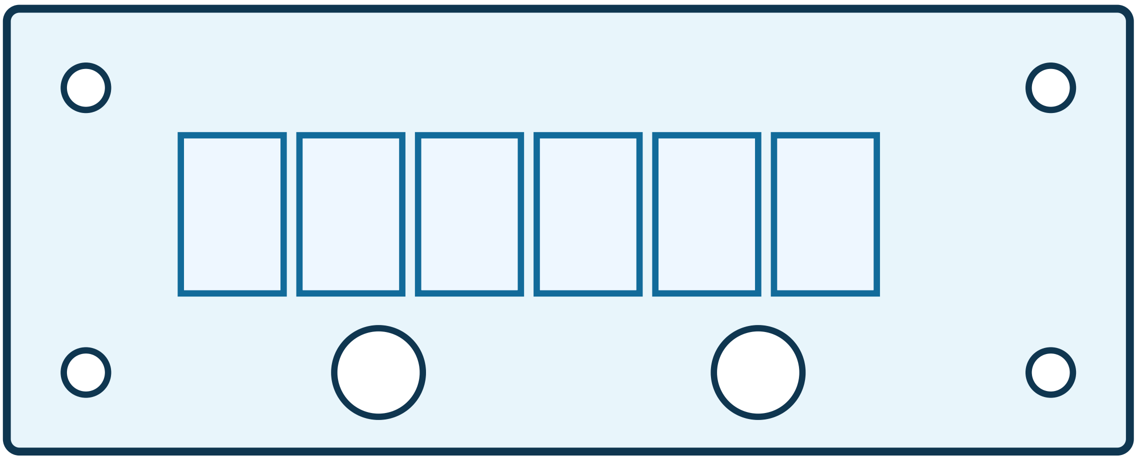
1. Create layer set and assign line types.
2. Draw 200 × 120 outer profile and centre lines.
3. OFFSET 20 from edges to locate four hole centres; draw Ø14 holes.
4. Construct two 70 × 16 slots at the specified centre spacing.
5. Add centre marks, overall dimensions, hole note and slot dimensions.
6. Add title and inspect plot preview on A4 landscape.

Plate
180 × 100

Terminal row
6 × 20 × 30

Fixing holes
4 × Ø8

Cable entries
2 × Ø24



TERMINAL MOUNTING PLATE

The completed sheet should show arrayed terminal modules, fixing holes, cable entries, dimensions and component labels.

1. Draw outer plate and inner construction margins.
2. Create one terminal block rectangle and use ARRAYRECT/COPY for six.
3. Locate four Ø8 fixing holes with offsets.
4. Create two Ø24 cable-entry holes on the lower centre line.
5. Add component numbers T1-T6, diameter notes and overall/spacing dimensions.
6. Prepare title block and plot preview.

Dimensioning and plotting checklist

Geometry

- ✓ Closed profiles
- ✓ No duplicate lines
- ✓ Exact centres and tangencies

Annotation

- ✓ No repeated dimensions
- ✓ Diameter/radius symbols correct

- ✓ Text height readable

Plot

- ✓ Correct paper size/orientation
- ✓ Line weights visible
- ✓ Drawing centred

Module 4 practice questions

PROCEDURE

Prepare a layer and dimension plan for the motor base plate.

APPLICATION

Explain how ARRAYRECT reduces work in the terminal plate.

QUALITY

List checks to perform before plotting a complete 2D drawing.

PRACTICAL

Complete either figure with title, dimensions and plot setup.

CAD COMMAND BANK

Purpose, syntax and practical use

Command names follow common AutoCAD terminology. Other CAD packages may use different icons or names, but the geometric operations remain similar.

LINE

Creates straight line segments.

```
LINE → specify first point → specify next point(s) → Enter
```

Practical tip: Use absolute, relative or polar coordinates. Each segment is a separate object.

Keyboard and workflow shortcuts

Key/action	Purpose	Lab use
Esc	Cancel current command.	Recover from an incorrect option safely.
Enter / Space	Accept option or repeat last command.	Faster repeated operations.
F3	Toggle OSNAP.	Precision point selection.
F8	Toggle ORTHO.	Horizontal/vertical drawing.
F10	Toggle POLAR.	Angle tracking.
F5 / Ctrl+E	Cycle isoplanes.	Isometric drawing.
Ctrl+S	Save.	Protect work frequently.
Ctrl+Z	Undo.	Reverse the last action.

File safety
Save after setup, after main geometry, after annotation and before plotting. Keep a backup copy. Do not overwrite another student's file.

LAB WORKBOOK

Record format, submission checklist and self-assessment

Recommended record format for every exercise

Record item	What to write
Experiment number and title	Example: M2-F3 - Tangent cam profile.
Aim	One sentence stating the required CAD drawing.
Software/tools	CAD package and important commands.
Given data	Dimensions, quantities and assumptions.
Procedure	Ordered command and construction steps.
Result	Drawing completed, checked, saved and plotted/exported.
File evidence	DWG/DXF filename and screenshot/print.
Learning reflection	One error faced and how it was corrected.

Minimum semester drawing register

Course 2038

No.	Exercise	Official minimum category	File name	Checked
1	Starter plate with four holes	M1 sketch	2038_M1_Starter.dwg	☐
2	Slotted mounting plate	M2 Figure 1	2038_M2_F1.dwg	☐
3	Circular flange	M2 Figure 2	2038_M2_F2.dwg	☐
4	Tangent cam	M2 Figure 3	2038_M2_F3.dwg	☐
5	Stepped support	M2 Figure 4	2038_M2_F4.dwg	☐
6	Keyhole plate	M2 Figure 5	2038_M2_F5.dwg	☐
7	Panel mounting layout	M2 Figure 6	2038_M2_F6.dwg	☐
8	Block with hole	M3 Isometric 1	2038_M3_I1.dwg	☐
9	Stepped block	M3 Isometric 2	2038_M3_I2.dwg	☐
10	U bracket	M3 Isometric 3	2038_M3_I3.dwg	☐
11	Electric motor form	M3 Isometric 4	2038_M3_I4.dwg	☐
12	Motor base plate	M4 2D Figure 1	2038_M4_D1.dwg	☐
13	Terminal mounting plate	M4 2D Figure 2	2038_M4_D2.dwg	☐

Drawing quality checklist

Accuracy

- ✓ All specified dimensions used
- ✓ No visual guessing
- ✓ Centres and tangencies exact
- ✓ Closed profiles where required

CAD method

- ✓ Correct commands selected
- ✓ Layers used logically
- ✓ No unnecessary duplicate entities
- ✓ File saved with correct name

Presentation

- ✓ Dimensions readable
- ✓ Line types visible
- ✓ Title and labels present

✓ Plot preview checked
Course 2038

Viva practice

VIVA
Why is OSNAP more reliable than visual selection?

VIVA
Differentiate OFFSET and COPY.

VIVA
Why should dimensions be placed on a separate layer?

VIVA
What is the purpose of plot preview?

VIVA
How is an isocircle created?

VIVA
What happens if the wrong units are used?

OFFICIAL EVALUATION TIMING

Two model three-hour lab examinations

The syllabus allocates Lab Exam 1 after Module 2 and Lab Exam 2 after Module 4, three hours each. The following are fresh practice models, not an official mark scheme.

Task: Draw a 160 × 100 mounting plate with R12 rounded corners, four Ø12 holes located 20 mm from adjacent edges, a central Ø40 opening, and two horizontal slots 50 × 12 placed symmetrically above and below the centre. Use separate object, centre and dimension layers. Add all essential dimensions and prepare A4 landscape plot preview.

Expected submission

- ✓ DWG/DXF source file
- ✓ Correct layer structure
- ✓ Fully dimensioned drawing
- ✓ Plot preview or PDF

Planning
Units, file and layers set before drawing.

Geometry
Dimensions, centres, radii and slots accurate.

Technique
Appropriate use of OFFSET, COPY/MIRROR, FILLET and TRIM.

Presentation
Readable annotation and plot setup.

Task: Create an isometric drawing of a U-shaped equipment support having an overall base 100 × 60 × 12, two side walls 12 thick and 70 high, and one Ø20 through hole in each side wall at equal height. Add a suitable title and prepare plot preview.

Expected submission

- ✓ Correct isometric snap and isoplane use
- ✓ Base and equal wall thickness
- ✓ Two correctly oriented isocircles
- ✓ Clean visible outline and title

Visualisation
Correct interpretation of dimensions.

Isometric method
Axes, isoplanes and isocircles used correctly.

Accuracy
Constant thickness and aligned holes.

Output
Saved file and clear plot preview.

Lab exam strategy

1. Spend the first 10 minutes reading, listing dimensions and planning commands.
2. Set units, save and create layers before geometry.
3. Create construction geometry first; do not dimension incomplete geometry.
4. Use symmetry, arrays and offsets to reduce repeated manual work.
5. Reserve the final 25 minutes for checking, annotation, title, save and plot preview.

STUDENT ACTIVITY • CONTINUOUS INTERNAL EVALUATION

Open-ended group experiment

The official syllabus specifies groups of 3-5, no duplication between groups, a separate report and 15 marks. The given example is an isometric drawing of an electric machine using CAD software.

Official conditions
Work in a group of 3-5. Select a unique experiment not duplicated by another group. Prepare and submit a separate report. The activity is intended for continuous internal evaluation with a score of 15 marks.

Suggested project themes

Electric machine

Develop a simplified isometric drawing of a motor or generator with base, body, shaft and terminal box.

Renewable-energy frame

Model a simplified solar-panel mounting frame or battery rack.

Electrical panel

Create a 2D panel layout with device rows, cable entries and mounting dimensions.

EV component support

Develop a mounting bracket or battery-module tray using 2D and isometric views.

Workshop fixture

Draw a simple jig, clamp or equipment support.

Campus utility

Prepare a dimensioned equipment-base or cable-tray support drawing.

Project workflow

1. Choose a unique object and obtain instructor approval.
2. Measure or define safe simplified dimensions.
3. Prepare a freehand concept sketch and drawing plan.
4. Assign group roles: geometry, layers/annotation, verification, report and presentation.
5. Create CAD drawing, review dimensions and produce plot/PDF.
6. Write the report and present the method, difficulties and corrections.

Separate report structure

Section	Required content
Title page	Course, project title, group members, register numbers and date.
Objective	What the group intended to draft/model.
Object description	Function, main parts and simplified assumptions.
Dimensions/data	Dimension table and source of data.
CAD plan	Layers, commands and drawing sequence.
Procedure	Step-by-step execution with screenshots.
Final output	Dimensioned 2D/isometric drawing and plot.
Challenges	Errors, corrections and team learning.
Conclusion	Result and possible improvement.

Suggested 15-mark rubric (instructor adaptable)

Planning and uniqueness: 3; CAD accuracy: 5; use of tools/layers: 3; report: 2; teamwork/presentation: 2. This is a suggested handbook rubric, not an official distribution stated in the syllabus.

MASTER ANSWER KEY

Complete answers and procedure-based solutions

This section contains answers to module questions, viva questions and full model lab-exam procedures.

Module 1 answers

Model space vs layout space: Model space is used to create full-size geometry. Layout space represents the paper sheet and viewports used for plotting at a selected scale.

75 mm at 30°: Enter `@75<30` after selecting the start point.

Tangent line between circles: Enable Tangent OSNAP and select the first circle at the tangent marker, then the second circle at its tangent marker.

Metric startup: New drawing → UNITS Decimal/mm → SAVEAS → create layers → set line types → set OSNAP/ORTHO/POLAR → begin construction.

Gaps after zooming: Likely causes are visual clicking without OSNAP, wrong snap mode, incomplete TRIM/EXTEND or duplicate objects. Turn on Endpoint/Intersection snaps, use EXTEND/TRIM, and inspect with zoom.

Module 2 answers

Four-hole flange procedure

1. Set units and layers.
2. Draw concentric circles Ø120, Ø80 and Ø40.
3. Use the Ø80 circle as a construction pitch circle.
4. At one quadrant, draw a Ø12 bolt hole.
5. ARRAYPOLAR around the common centre with 4 items and 360° fill.
6. Add centre marks and diameter/PCD note.
7. Freeze/delete construction circle only if required by the instructor.

Array rotates holes incorrectly: Check the array centre, item count, fill angle, association option and whether “rotate items” affects a non-circular object. For circular holes, centre location and item count are the critical settings.

Why Tangent OSNAP: It computes the exact contact point where a line touches a circle without crossing it. Visual placement cannot guarantee tangency.

OFFSET vs COPY vs ARRAY: OFFSET creates a parallel/concentric object at a fixed distance; COPY duplicates selected objects using displacement; ARRAY produces multiple copies in a rectangular, polar or path pattern.

A4 plot preparation: Open layout/plot, select A4 and orientation, choose printer/PDF, select plot area/layout, centre the drawing, set scale or Fit only when appropriate, apply line-weight plot style, preview, then plot.

Module 3 answers

Three isoplanes: Left, Top and Right.

Circle on top face: Activate isometric snap, switch to Top isoplane, run ELLIPSE, select Isocircle, choose centre and enter radius/diameter.

Cylindrical object procedure

1. Select the isoplane perpendicular to the cylinder axis.
2. Draw the near-end isocircle.
3. Project the cylinder length along the correct isometric axis.
4. Copy/draw the far-end isocircle.
5. Join corresponding extreme tangent points.
6. Trim or suppress hidden portions according to the required convention.

Wrong isocircle plane: Undo/delete it, cycle to the correct isoplane with F5/Ctrl+E, then recreate the isocircle.

Bounding box method: First draw the maximum overall length, width and height. Mark cuts/steps on the box edges, join corresponding points, then remove construction edges. This preserves proportions and axis directions.

Module 4 answers

Layer plan: OBJECT continuous/thick; CENTRE centre line/thin; HIDDEN dashed/thin when needed; DIMENSION continuous/thin; CONSTRUCTION light; TEXT continuous. Draw geometry first, then dimensions.

ARRAYRECT in terminal plate: It creates equally spaced copies in rows/columns from one accurate terminal rectangle, reducing repeated measurement and maintaining uniform spacing.

Pre-plot checks: Units, overall size, duplicate/gapped entities, layer assignment, dimension completeness, text height, line-type scale, line weights, paper size, orientation, plot area, scale and preview.

Viva answers

Question	Model answer
Why OSNAP?	It selects exact endpoints, centres, intersections, quadrants and tangencies, ensuring geometric accuracy.
OFFSET vs COPY?	OFFSET creates parallel/concentric geometry at a distance; COPY produces a duplicate at a chosen displacement.
Separate dimension layer?	Dimensions can be controlled, hidden, printed and edited independently from object geometry.
Purpose of plot preview?	It reveals clipping, scale, orientation, line-weight and readability problems before final output.
How is an isocircle created?	Isometric snap → correct isoplane → ELLIPSE → Isocircle.
Wrong units effect?	All entered distances and scale relationships become incorrect, causing an unusable engineering drawing.

Complete Model Lab Exam 1 solution

1. Start metric drawing; save as `2038_LabExam1.dwg`.
2. Create OBJECT, CENTRE, DIMENSION and CONSTRUCTION layers.
3. RECTANGLE from 0,0 to @160,100. FILLET radius 12 on all corners.
4. OFFSET 20 inward from left/right/top/bottom construction references. Their four intersections are hole centres.
5. Draw one CIRCLE diameter 12 and COPY to the other three centres.
6. Draw centre lines through plate centre. Draw central CIRCLE diameter 40.
7. For the upper slot, mark two centres 50 mm apart on a horizontal line. Draw two R6 circles, join with tangent horizontal lines and TRIM inner arcs. Place the slot symmetrically above centre.
8. MIRROR or COPY the slot to an equal distance below centre.
9. Remove unnecessary construction lines; apply centre marks.
10. Dimension overall size, R12, 4×Ø12, Ø40, slot 50×12 and location/spacing dimensions.
11. Add title, save, open A4 landscape plot preview and check line weights/readability.

Expected final view: one symmetric rounded rectangular plate with four corner holes, central circular opening and two equal horizontal slots.

Complete Model Lab Exam 2 solution

1. Start metric drawing, save as `2038_LabExam2.dwg`, enable isometric snap.
2. Draw the 100 × 60 top outline of the base along isometric axes.
3. Project 12 mm downward and close the base slab.
4. On the left and right edges, construct 12 mm wall thickness.
5. From the top of the base, project both walls to 70 mm height and close their top faces.
6. Mark equal hole centres on both walls using offsets from base and wall edges.
7. Switch to the correct left/right isoplane and create Ø20 isocircles.
8. Trim construction lines and retain clean visible outlines.
9. Add a title; optional dimensions may be placed if required by the instructor.
10. Save and prepare plot preview.

Expected final view: a symmetric isometric U bracket with a 12 mm base, equal 12 mm side walls and aligned circular holes represented by correctly oriented isocircles.

OFFICIAL REFERENCES

Textbooks and online resources from the syllabus

Code	Book / author
T1	*Engineering Graphics with Auto CAD* - R. B. Choudary, Anuradha Publishers.
T2	K. C. John, *Engineering Graphics*, PHI Learning Private Limited.
R1	P. I. Varghese, *Engineering Graphics*, VIP Publishers.
R2	Anilkumar K. N., *Engineering Graphics*, Adhyuth Narayan Publishers.
R3	N. D. Bhatt, *Engineering Drawing*, Charotar Publishing House, Anand, Gujarat, 2010; ISBN 978-93-80358-17-8.
R4	M. B. Shah and B. C. Rana, *Engineering Drawing*, Pearson Publications.

Online resources

No.	Resource	How it supports the course
1	AutoCAD overview	General software background and terminology.
2	NPTEL course 112102101	Engineering drawing/CAD learning support.
3	YouTube CAD tutorial	Video demonstration.
4	Introduction to AutoCAD PDF	Introductory command and interface reference.
5	AutoCAD tutorial PDF page	Additional guided practice.
6	AutoCAD Tutorial 002	Supplementary tutorial material.

Software note

Interfaces and command options vary by software version. Use the installed laboratory version and follow instructor guidance. The handbook focuses on stable geometric methods rather than one specific screen layout.

Final practical checklist

Before drawing

- ✓ Question read completely
- ✓ Units set
- ✓ File saved
- ✓ Layers created
- ✓ OSNAP selected

Before submission

- ✓ Geometry checked
- ✓ No duplicates/gaps
- ✓ Dimensions complete
- ✓ Title present
- ✓ Plot preview checked
- ✓ Final save completed

Engineering Graphics Using CAD Software • Course 2038
Author: Nandakumar.M | Visit: nandakumarm.dpdns.org
Standalone practical handbook prepared from the official syllabus.